

Maximum Likelihood Estimator (MLE)

Elementary introduction

Consider Bernoulli trials with a coin which has $P(H)=P(X=1)=\theta$. Suppose the outcome of three tosses is {HTH}. The probability of this outcome is $P(X_1=1, X_2=0, X_3=1)=\theta^2(1-\theta)$, which is 1/8 when $\theta=1/2$. A joint pdf $f(x_1, \dots, x_n; \theta)$ of a random sample thus tells us the probability of a set of specific observations, *given the distribution parameters*. That is, a joint pdf is treated as a function of the values of x_i , given θ .

In estimation problems, we do not know the distribution parameters, but have a set of observations. The question we pose is: Which distribution parameter would have generated the sample on hand *most likely*? In the example above, is the observed sample {HTH} generated more likely by $\theta=1/2$, or by $\theta=2/3$? The probability of the observed sample is 1/8 if $\theta=1/2$, and 4/27 if $\theta=2/3$. Therefore, it is more likely that the observed sample is generated by $\theta=2/3$. In fact, $\theta=2/3$ is the value that maximizes the joint probability $\theta^2(1-\theta)$ of the observations {HTH}. In this estimation method, the joint pdf is treated as a function of θ , $f(\theta; x_1, \dots, x_n)$, given a sample observation $(x_1, \dots, x_n)$. Interpreted this way, we call the joint pdf a *likelihood function*.

The method of maximum likelihood estimation proposed by R.A. Fisher finds the estimators of population parameters by maximizing the likelihood function or the log-likelihood function with respect to θ . The resulting estimator is called the *maximum likelihood estimator* (MLE). Since the logarithmic function is a monotonically increasing function, maximization of likelihood function and log-likelihood function should yield the same solution. Note that the MLE may not exist, or if it exists, it may not be unique. The MLE requires the knowledge of the population distribution function to specify the likelihood function while the least squares method does not.

A Simple Example

Let $X_i, i=1, \dots, n$, be the random sample from a Bernoulli distribution $f(x; \theta) = \theta^x(1-\theta)^{1-x}$. Since X_i 's are i.i.d., the likelihood function L is the multiplication of n Bernoulli densities:

$$L(\theta; x_1, \dots, x_n) = \prod_{i=1}^n \theta^{x_i}(1-\theta)^{1-x_i} = \theta^{\sum x_i} (1-\theta)^{n-\sum x_i} = \theta^{n\bar{x}} (1-\theta)^{n(1-\bar{x})}$$

and the log-likelihood function

$$\ln L(\theta; x_1, \dots, x_n) = n\bar{x} \ln \theta + n(1-\bar{x}) \ln(1-\theta)$$

Taking the derivative of $\ln L$ with respect to θ , we find

$$\frac{\partial \ln L}{\partial \theta} = \frac{n\bar{x}}{\theta} - \frac{n(1-\bar{x})}{1-\theta} = 0 \Rightarrow \hat{\theta} = \bar{x}$$

$$\frac{\partial^2 \ln L}{\partial \theta^2} = -\frac{n\bar{x}}{\theta^2} - \frac{n(1-\bar{x})}{(1-\theta)^2} < 0 \quad \because \quad 0 \leq \bar{x} \leq 1$$

To consider the MLE more formally, let $\mathbf{y}' = (y_1, y_2, \dots, y_N)$ be a random sample of size N from a population characterized by a density $f(\mathbf{y}; \boldsymbol{\theta})$, where $\boldsymbol{\theta}$ is a K -dimensional vector in a compact parameter space Θ .

Definition: Likelihood function L and log likelihood function ℓ

$$L(\mathbf{y}; \boldsymbol{\theta}) = \prod_{i=1}^N f(y_i; \boldsymbol{\theta}), \quad \ell(\mathbf{y}; \boldsymbol{\theta}) = \ln L(\mathbf{y}; \boldsymbol{\theta}) = \sum_{i=1}^N \ln f(y_i; \boldsymbol{\theta}) = \sum_{i=1}^N \ell(y_i; \boldsymbol{\theta})$$

Definition: $\hat{\boldsymbol{\theta}}$ is the global MLE of the true parameter $\boldsymbol{\theta}_0$ if $\ell(\mathbf{y}; \hat{\boldsymbol{\theta}}) \geq \ell(\mathbf{y}; \boldsymbol{\theta})$ for all $\boldsymbol{\theta} \in \Theta$, and $\hat{\boldsymbol{\theta}}$ is the local MLE of $\boldsymbol{\theta}_0$ if it is a root of the gradient $\partial \ell / \partial \boldsymbol{\theta} = 0$.

We consider only the local MLE or consider the case of likelihood function which has a unique global maximum point. In this case, we find the MLE from the first order condition

$$\mathbf{g}(\boldsymbol{\theta}) = \frac{\partial \ln L}{\partial \boldsymbol{\theta}} = 0, \quad \mathcal{H}(\boldsymbol{\theta}) = \frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} \text{ negative definite}$$

Theorem. The MLE is a sufficient statistic and a best asymptotically normal (BAN) estimator. More importantly, the MLE has a *invariance property*, that is, if $\hat{\boldsymbol{\theta}}$ is the MLE of $\boldsymbol{\theta}$, then $\mathbf{h}(\hat{\boldsymbol{\theta}})$ is the MLE of $\mathbf{h}(\boldsymbol{\theta})$ for any continuous function $\mathbf{h}$.

Regularity Condition: The joint pdf $L(\mathbf{y}; \boldsymbol{\theta})$ is said to be regular if

- (a) The sample space of $\mathbf{y}$ for which $L(\mathbf{y}; \boldsymbol{\theta})$ is strictly positive does not depend on $\boldsymbol{\theta}$.
- (b) For all $\boldsymbol{\theta} \in \Theta$ and for all $\mathbf{y}$, $L(\mathbf{y}; \boldsymbol{\theta})$ and $\ell(\mathbf{y}; \boldsymbol{\theta})$ have finite-valued partial derivatives with respect to $\boldsymbol{\theta}$ up to the third order.
- (c) The covariance matrix of $\partial \ln L / \partial \boldsymbol{\theta}$ is positive definite everywhere in Θ .
- (d) The integral $\int \ln L(\mathbf{y}; \boldsymbol{\theta}) d\mathbf{y}$ is twice differentiable under the integral.

Theorem. Asymptotic Properties of MLE. Let $\hat{\boldsymbol{\theta}}$ be the MLE of the true parameter $\boldsymbol{\theta}_0$ for a sample of size N . Then, under the regularity conditions, $\hat{\boldsymbol{\theta}}$ is consistent and has an asymptotic normal distribution, i.e.,

$$(a) \text{plim}_{N \rightarrow \infty} \hat{\boldsymbol{\theta}} = \boldsymbol{\theta}_0$$

$$(b) \sqrt{N}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{\Psi}^{-1}), \quad \boldsymbol{\Psi} = E\left(-\frac{1}{N} \mathcal{H}(\boldsymbol{\theta})\right)$$

where $\boldsymbol{\Psi}$ is Fisher's *information matrix*.

Alternative computational forms of the information matrix are

$$\begin{aligned}\Psi &= E\left(-\frac{1}{N}\mathcal{H}(\theta)\right) = E\left(\frac{1}{N}\mathbf{g}(\theta)\mathbf{g}(\theta)'\right) = \text{plim}\left(-\frac{1}{N}\mathcal{H}(\theta)\right) \\ &= E[-\mathcal{H}_t(\theta)] = E[\mathbf{g}_t(\theta)\mathbf{g}_t(\theta)'] = \text{plim}\left(\frac{1}{N}\sum_{t=1}^N \mathbf{g}_t(\theta)\mathbf{g}_t(\theta)'\right)\end{aligned}$$

where subscript t indicates a single observation

$$\mathbf{g}_t(\theta) = \frac{\partial \ln f(\mathbf{y}_t; \theta)}{\partial \theta}, \quad \mathcal{H}_t(\theta) = \frac{\partial^2 \ln f(\mathbf{y}_t; \theta)}{\partial \theta \partial \theta'} = \frac{\partial \mathbf{g}_t(\theta)}{\partial \theta'}$$

Note that $\mathbf{g}(\theta) = \sum \mathbf{g}_t(\theta)$ and $\mathcal{H}(\theta) = \sum \mathcal{H}_t(\theta)$. Note also that, since Ψ^{-1} is the covariance matrix of $\sqrt{N}\hat{\theta}$, the covariance matrix of $\hat{\theta}$ is computed by $(N\Psi)^{-1}$.

The proof of the asymptotic properties of the MLE under the regularity conditions is based on the following propositions. Entire proofs of the theorem and propositions are given at the end of this note.

Proposition 1: $E[\ell(\theta_0)/N] \geq E[\ell(\theta)/N]$ for all $\theta \in \Theta$.

Proposition 2: $\text{plim}_{N \rightarrow \infty} \ell(\theta)/N = E[\ell(\theta)/N]$ for all $\theta \in \Theta$.

Proposition 3: $E[\mathbf{g}(\theta)] = E[\mathbf{g}_t(\theta)] = \mathbf{0}$, $\text{cov}[\mathbf{g}(\theta)] = N\Psi$, $\text{cov}[\mathbf{g}_t(\theta)] = \Psi$

Proposition 4: $\frac{1}{\sqrt{N}}\mathbf{g}(\theta) \xrightarrow{d} N(\mathbf{0}, \Psi)$

Proposition 5: $\text{plim}_{N \rightarrow \infty} \left(-\frac{1}{N}\mathcal{H}(\theta)\right) = \Psi$

Restricted Maximum Likelihood Estimator

Let $L(y; \theta)$ and $\ell(\theta) = \ln L(y; \theta)$ be as defined before, where θ is a K -dimensional vector of unknown parameters. The gradient vector and the Hessian matrix are denoted by $g(\theta)$ and $\mathcal{H}(\theta)$, respectively. Let $h(\theta) = 0$ be a J -dimensional vector of restrictions on θ . The restricted MLE of θ is obtained by maximizing the Lagrange equation

$$\mathcal{L} = \ell(\theta) - \lambda' h(\theta)$$

where λ is a J -dimensional vector of Lagrange multipliers. The restricted MLE $\tilde{\theta}$ and $\tilde{\lambda}$ are the solutions to the first order conditions

$$g(\tilde{\theta}) - H_{\tilde{\theta}} \lambda = 0$$

$$h(\tilde{\theta}) = 0$$

where $H_{\tilde{\theta}} = \partial h(\tilde{\theta}) / \partial \theta$ is a $K \times J$ matrix evaluated at $\tilde{\theta}$. The subscript for H emphasizes that it may be a function of θ .

Asymptotic Distribution of Restricted MLE

Let $H_0 = \partial h(\theta_0) / \partial \theta$, and let $\mathcal{H}_0$ and Ψ_0 be the Hessian matrix and the information matrix evaluated at the true parameter value θ_0 . Then, $\tilde{\theta}$ and $\tilde{\lambda}$ are asymptotically uncorrelated, and

$$\sqrt{N}(\tilde{\theta} - \theta_0) \xrightarrow{d} N(\mu_{\tilde{\theta}}, \Sigma_{\tilde{\theta}})$$

$$\frac{1}{\sqrt{N}} \tilde{\lambda} \xrightarrow{d} N(\mu_{\tilde{\lambda}}, \Sigma_{\tilde{\lambda}})$$

where, dropping zero subscripts for notational simplicity,

$$\begin{aligned} \mu_{\tilde{\theta}} &= \Psi^{-1} H (H' \Psi^{-1} H)^{-1} h(\theta_0), & \Sigma_{\tilde{\theta}} &= \Psi^{-1} - \Psi^{-1} H (H' \Psi^{-1} H)^{-1} H' \Psi^{-1} \\ \mu_{\tilde{\lambda}} &= -(H' \Psi^{-1} H)^{-1} h(\theta_0), & \Sigma_{\tilde{\lambda}} &= (H' \Psi^{-1} H)^{-1} \end{aligned}$$

To show these results, linearize the first order conditions by Taylor's expansion around the true parameter θ_0 :

$$g(\theta_0) + \mathcal{H}_0(\tilde{\theta} - \theta_0) - H_0 \tilde{\lambda} = 0$$

$$h(\theta_0) + H_0'(\tilde{\theta} - \theta_0) = 0$$

Note that $H_{\tilde{\theta}}$ in the first equation is simply replaced with H_0 , instead of expanding $H_{\tilde{\theta}}$ around θ_0 . See Silvey

(1970) for the reasons why this can be done.

Dividing the first equation by $\sqrt{N}$, multiplying the second equation by $\sqrt{N}$, and then rearranging the terms, we can write the two equations in a matrix form as

$$\begin{pmatrix} -\frac{1}{N}\mathfrak{H}_0 & H_0 \\ H_0' & 0 \end{pmatrix} \begin{pmatrix} \sqrt{N}(\tilde{\theta} - \theta_0) \\ \frac{1}{\sqrt{N}}\tilde{\lambda} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{N}}g(\theta_0) \\ -\sqrt{N}h(\theta_0) \end{pmatrix}$$

Therefore, the solution is given by

$$\tilde{\delta} \equiv \begin{pmatrix} \sqrt{N}(\tilde{\theta} - \theta_0) \\ \frac{1}{\sqrt{N}}\tilde{\lambda} \end{pmatrix} = \begin{pmatrix} -\frac{1}{N}\mathfrak{H}_0 & H_0 \\ H_0' & 0 \end{pmatrix}^{-1} \begin{pmatrix} \frac{1}{\sqrt{N}}g(\theta_0) \\ -\sqrt{N}h(\theta_0) \end{pmatrix}$$

Note that $h(\theta)=0$ if the restrictions are true. By Proposition 5, the first right hand side matrix converges in probability to

$$plim_{N \rightarrow \infty} \begin{pmatrix} -\frac{1}{N}\mathfrak{H}_0 & H_0 \\ H_0' & 0 \end{pmatrix}^{-1} = \begin{pmatrix} \Psi_0 & H_0 \\ H_0' & 0 \end{pmatrix}^{-1} \equiv Q$$

and by Proposition 4, the last right hand side vector converges in distribution to a normal distribution

$$\begin{pmatrix} \frac{1}{\sqrt{N}}g(\theta_0) \\ -\sqrt{N}h(\theta_0) \end{pmatrix} \xrightarrow{d} N(m, \Omega), \quad m \equiv \begin{pmatrix} 0 \\ -\sqrt{N}h(\theta_0) \end{pmatrix}, \quad \Omega \equiv \begin{pmatrix} \Psi_0 & 0 \\ 0 & 0 \end{pmatrix}$$

Therefore,

$$\tilde{\delta} \xrightarrow{d} N(Qm, Q\Omega Q')$$

By using the formula for the inverse of a partitioned matrix, one can easily verify

$$Q = \begin{pmatrix} \Psi^{-1} - \Psi^{-1}H(H'\Psi^{-1}H)^{-1}H'\Psi^{-1} & \Psi^{-1}H(H'\Psi^{-1}H)^{-1} \\ (H'\Psi^{-1}H)^{-1}H'\Psi^{-1} & -(H'\Psi^{-1}H)^{-1} \end{pmatrix}$$

where the subscripts for Ψ and H are dropped for notational convenience. This gives

$$Qm = \begin{pmatrix} \Psi^{-1}H(H'\Psi^{-1}H)^{-1}h(\theta_0) \\ -(H'\Psi^{-1}H)^{-1}h(\theta_0) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \text{if } h(\theta_0) = 0$$

$$Q\Omega Q' = \begin{pmatrix} \Psi^{-1} - \Psi^{-1}H(H'\Psi^{-1}H)^{-1}H'\Psi^{-1} & 0 \\ 0 & (H'\Psi^{-1}H)^{-1} \end{pmatrix}$$

This covariance matrix indicates that

- (i) the covariance matrix for $\sqrt{N}(\tilde{\theta} - \theta_0)$ is singular with rank (K-J),
- (ii) the restricted estimator of θ has a smaller covariance matrix than the covariance matrix Ψ^{-1} of the unrestricted estimator, and
- (iii) the restricted estimator of θ and the estimator of Lagrange multiplier are asymptotically independent.

Consistency of $\tilde{\theta}$ and $\tilde{\lambda}$ can also be shown in a similar way. Note that they are not consistent if the restrictions are false ($h(\theta) \neq 0$).

Asymptotic Tests based on the MLE

We have already introduced the basic idea of the *likelihood ratio test*, the *Wald test*, and the *Lagrange Multiplier test* (*Efficient Score test*, χ^2 test) in the previous discussion of finite sample tests based on the least squares estimators. Here, we recap the original ideas based on the maximum likelihood estimators. These tests lead to asymptotic χ^2 distributions. The null and alternative hypotheses are composite hypotheses of two sided test

$$H_0: h(\theta)=0, \quad H_1: h(\theta) \neq 0$$

(1) *Likelihood Ratio Test (LRT)*

Recall that the MLE maximizes the log likelihood function. We know $\ln L(\hat{\theta})$ is greater than $\ln L(\tilde{\theta})$. What can we say about the magnitude of the difference? If the restrictions imposed by the null hypothesis are true, maximization with and without restrictions would not make much difference, and we expect the likelihood values are close to each other. On the other hand, if the restrictions imposed by the null hypothesis are not true and they are really binding restrictions, then we expect a large difference between the two likelihood values. This suggests that we may use the difference $\ln L(\hat{\theta}) - \ln L(\tilde{\theta})$ as the test statistic. The likelihood ratio test uses two times of this difference as the test statistic

$$LRT = 2[\ln L(\hat{\theta}) - \ln L(\tilde{\theta})]$$

Once the test statistic is chosen, we need to determine the decision rule of when to reject H_0 in favor of H_1 . As alluded to already, we will reject H_0 if LRT is *sufficiently large*. Otherwise, the test concludes that the data does not provide enough evidence to reject H_0 . How big a difference is sufficiently large enough to reject the null hypothesis? This requires a choice of the critical (rejection) region of the test. Since we reject the null hypothesis when LRT is large, the proper critical region is the set of the sample, $R_c = \{LRT > c\}$, where c is a pre-chosen critical value. The choice of the critical value depends on how much of Type I error we are willing to tolerate. A choice of large critical value will reject the null hypothesis less frequently and we are more likely to commit the Type I error. The choice of the critical value is governed by the choice of *significance level* (*size of the test*, or *test size*). That is, for a pre-chosen significance level α , the critical value is determined by

$$P(LRT > c \mid H_0) = \alpha$$

which requires the knowledge of the distribution function of the test statistic LRT under the null hypothesis. The following theorem provides the answer.

Theorem. Let $\hat{\theta}$ and $\tilde{\theta}$ be the unrestricted and restricted MLE of θ , respectively, and let J be the number of restrictions under $H_0: h(\theta)=0$. Under the regularity conditions, the likelihood ratio test statistic has an asymptotic central $\chi^2(J)$ under H_0

$$LRT \equiv 2 \ln \Lambda = 2[\ln L(\hat{\theta}) - \ln L(\tilde{\theta})] \xrightarrow{d} \chi^2(J) \text{ under } H_0$$

The LRT has a noncentral χ^2 distribution under the alternative hypothesis.

Proof. The proof is in the Appendix.

The critical value for the LRT is found from $P(\chi^2(J) \leq c) = 1 - \alpha$. For example, the critical value for 5% significance level ($\alpha=0.05$) is 5.99 for $J=2$, and we reject the null hypothesis if $LRT > 5.99$. The probability of rejecting correct null hypothesis in this example is 0.05. That is, if we repeatedly apply this test to many different samples, we make a mistake of rejecting the correct null hypothesis 5% of the cases.

(2) Wald Test

Unlike the comparison of the unrestricted and restricted likelihoods in the LRT test, the Wald test compares the values of the restriction equation evaluated at unrestricted and restricted estimators. Since $h(\tilde{\theta})=0$, this is the same as examining whether the unrestricted estimator satisfies the restrictions of H_0 or not, i.e., whether $h(\hat{\theta})$ is close to the hypothesized value of zero or not. The null hypothesis is rejected if $h(\hat{\theta})$ is not sufficiently close to zero. Two features of the Wald test are to be noted:

- (i) Since the test is based on the unrestricted value $h(\hat{\theta})$, only unrestricted estimators enter the test statistic.
- (ii) Unlike the scalar value of the likelihood function, $h(\hat{\theta})$ is a $J \times 1$ vector in general. Therefore, we need to choose a measure the distance of $h(\hat{\theta})$ from zero. One may just consider the squared Euclidean norm, $h(\hat{\theta})' h(\hat{\theta})$, i.e., the sum of squared values of each restriction $h_i(\hat{\theta})$. However, this does not take into account the differences in the variances $h_i(\hat{\theta})$. We need to normalize $h(\hat{\theta})$ in a certain way.

Let $H = \partial h(\theta_0) / \partial \theta$ evaluated at the true value θ_0 and Ψ be the information matrix. The Wald test statistic is defined by the quadratic form of $h(\hat{\theta})$

$$WT \equiv N h(\hat{\theta})' [H' \Psi^{-1} H]^{-1} h(\hat{\theta}) \xrightarrow{d} \chi^2(J) \text{ under } H_0$$

This test statistic has an asymptotic central $\chi^2(J)$ under the null hypothesis, and has an asymptotic noncentral $\chi^2(J, \delta)$ under the alternative hypothesis $h(\theta_0) \neq 0$ with the noncentrality parameter

$$\delta^2 = h(\theta_0)' [H' \Psi^{-1} H]^{-1} h(\theta_0)$$

Matrices H and Ψ will be evaluated by the unrestricted estimate $\hat{\theta}$. We reject the null hypothesis if WT is greater than the critical value for a chosen significance level.

Theorem. The Wald test statistic WT has an asymptotic χ^2 distribution.

Proof. We have shown the asymptotic normality of the MLE

$$\sqrt{N}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \Psi^{-1})$$

To derive the asymptotic distribution of $h(\hat{\theta})$, we use the *delta method* by taking the first order Taylor

expansion of $h(\hat{\theta})$ around the true value θ_0

$$h(\hat{\theta}) \approx h(\theta_0) + H'(\hat{\theta} - \theta_0)$$

Rearranging the terms, we can write

$$\sqrt{N}[h(\hat{\theta}) - h(\theta_0)] = H'[\sqrt{N}(\hat{\theta} - \theta_0)] \xrightarrow{d} N(0, H'\Psi^{-1}H)$$

where the last statement of asymptotic normal distribution is due to the asymptotic normal distribution of $\sqrt{N}(\hat{\theta} - \theta_0)$. Therefore, the quadratic form has a central χ^2 distribution

$$N[h(\hat{\theta}) - h(\theta_0)]' [H'\Psi^{-1}H]^{-1} [h(\hat{\theta}) - h(\theta_0)] \xrightarrow{d} \chi^2(J)$$

The null and non-null distributions of WT follow from this result.

q.e.d.

(3) *Lagrange Multiplier Test (Rao's Score Test)*

As the name indicates, this test is based on the restricted estimator $\tilde{\lambda}$ of the Lagrange multiplier λ in the estimation of restricted MLE. If the restriction $h(\theta)=0$ is true, it will not be binding constraints in the maximization of the likelihood function, and hence, the restricted estimator $\tilde{\lambda}$ (estimator of the shadow price of the restrictions) will be close to zero. Thus, we reject the null hypothesis if $\tilde{\lambda}$ is sufficiently different from a zero vector. Note that this test may be interpreted as a comparison of restricted and unrestricted estimators of λ because the unrestricted estimator $\hat{\lambda}$ is identically zero. As in the Wald test, we note

- (i) Since the test is based on the restricted estimator $\tilde{\lambda}$, only restricted estimators enter the test statistic.
- (ii) Since $\tilde{\lambda}$ is a vector in general, we need to find a measure of the distance between $\tilde{\lambda}$ and a zero vector, and need to normalize it.

We have shown earlier that $\tilde{\lambda}$ has an asymptotic normal distribution

$$\frac{1}{\sqrt{N}}\tilde{\lambda} \xrightarrow{d} N\left(-[H'\Psi^{-1}H]^{-1}h(\theta_0), [H'\Psi^{-1}H]^{-1}\right)$$

whose mean will be zero if $h(\theta_0)=0$. Therefore, the LM test statistic is defined as

$$LMT = \frac{1}{N}\tilde{\lambda}' [H'\Psi^{-1}H] \tilde{\lambda} \xrightarrow{d} \chi^2(J) \text{ under } H_0$$

i.e., it has an asymptotic central $\chi^2(J)$ under the null hypothesis, and has an asymptotic noncentral $\chi^2(J, \delta)$ under the alternative hypothesis $h(\theta_0) \neq 0$ with the noncentrality parameter $\delta^2 = h(\theta_0)' [H'\Psi^{-1}H]^{-1} h(\theta_0)$. We evaluate H and Ψ at the restricted estimate $\tilde{\theta}$, and reject the null hypothesis if LMT is greater than the critical value for a chosen significance level.

An alternative approach that leads to the same test statistic as the LM test considers the gradient vector $g(\theta)$. We know $g(\hat{\theta})=0$ because the unrestricted estimator $\hat{\theta}$ is the solution to $g(\theta)=0$. If the null hypothesis is true, we would expect that $g(\tilde{\theta})$ is also close to zero as the unrestricted and restricted estimators are close each other. The test based on the gradient vector is called the *score test*. It is easy to see

the equivalence of the score test and the LM test. The first order condition for the restricted maximum likelihood estimation is $\mathbf{g}(\tilde{\boldsymbol{\theta}}) = \mathbf{H}\tilde{\boldsymbol{\lambda}}$. Since matrix $\mathbf{H}$ is assumed to have a full column rank, $\mathbf{g}(\tilde{\boldsymbol{\theta}}) = \mathbf{0}$ if and only if $\tilde{\boldsymbol{\lambda}} = \mathbf{0}$, and hence, testing the closeness of $\tilde{\boldsymbol{\lambda}}$ to zero is the same as testing the closeness of $\mathbf{g}(\tilde{\boldsymbol{\theta}})$ to zero. Substituting $\mathbf{H}\tilde{\boldsymbol{\lambda}} = \mathbf{0}$ in the LM test statistic with $\mathbf{g}(\tilde{\boldsymbol{\theta}})$, we can rewrite the LMT as

$$LMT = \frac{1}{N} \tilde{\boldsymbol{\lambda}}' (\mathbf{H}' \boldsymbol{\Psi}^{-1} \mathbf{H}) \tilde{\boldsymbol{\lambda}} = \frac{1}{N} (\mathbf{H} \tilde{\boldsymbol{\lambda}})' \boldsymbol{\Psi}^{-1} (\mathbf{H} \tilde{\boldsymbol{\lambda}}) = \frac{1}{N} \mathbf{g}(\tilde{\boldsymbol{\theta}})' \boldsymbol{\Psi}^{-1} \mathbf{g}(\tilde{\boldsymbol{\theta}})$$

The last expression is the LM test in the score form. The LM test thus does not require the estimate of the Lagrange multiplier if one uses this score form of the LM test.

Remark: We have shown in the discussion of the asymptotic distribution of restricted MLE that $\mathbf{g}(\boldsymbol{\theta}_0)$ has a normal distribution

$$\frac{1}{\sqrt{N}} \mathbf{g}(\boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{\Psi}_0)$$

Therefore, the score test is a quadratic form of this normal vector and terms are evaluated at the restricted estimators.

Maximum Likelihood Estimator of a Standard Linear Regression Model

Consider a standard linear regression model with the assumption of normal error term:

$$y_t = X_t \beta + u_t \quad t=1,2,\dots,N, \quad u_t \sim NID(0, \sigma^2), \quad E(u_t X_t) = 0$$

Since the Jacobian of this transformation is 1, the pdf of y_t is

$$f(y_t; X_t, \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-(y_t - X_t \beta)^2 / (2\sigma^2)\right], \quad \theta = \begin{pmatrix} \beta \\ \sigma^2 \end{pmatrix}$$

and the likelihood function and log likelihood function are

$$L(\theta; y, X) = f(y; X, \theta) = \prod_{t=1}^N f(y_t; X_t, \theta) = (2\pi\sigma^2)^{-N/2} \exp\left(-\sum_{t=1}^N (y_t - X_t \beta)^2 / 2\sigma^2\right)$$

$$\ell(\theta) = -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\sigma^2) - \frac{(y - X\beta)'(y - X\beta)}{2\sigma^2}$$

The MLE of β and σ^2 are those that maximizes $\ell(\theta)$. The *gradient vector* (the *score*) is

$$\frac{\partial \ell(\theta)}{\partial \beta} = -\frac{1}{2\sigma^2} (-2X'y + 2X'X\beta) = \frac{1}{\sigma^2} (X'y - X'X\beta)$$

$$\frac{\partial \ell(\theta)}{\partial \sigma^2} = -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4} (y - X\beta)'(y - X\beta)$$

Setting the gradients to zero and solving for β and σ^2 give their MLE as

$$\hat{\beta} = (X'X)^{-1}X'y \quad \hat{\sigma}^2 = \frac{(y - X\hat{\beta})'(y - X\hat{\beta})}{N}$$

The MLE of β is identical to its OLSE, but the MLE of σ^2 is different from its OLSE. We know that $\hat{\beta}$ is unbiased, and $\hat{\sigma}^2$ is biased.

Substituting the MLE of β and σ^2 into the log likelihood function, we can find its maximum value as

$$\ell(\hat{\theta}) = -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\hat{\sigma}^2) - \frac{(y - X\hat{\beta})'(y - X\hat{\beta})}{2\hat{\sigma}^2} = -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\hat{\sigma}^2) - \frac{N}{2}$$

To find the asymptotic covariance matrix, we first find the Hessian matrix:

$$\mathcal{H} = (\mathcal{H}_{ij}) = \frac{\partial^2 \ell(\theta)}{\partial \theta \partial \theta'} = \begin{pmatrix} \frac{\partial^2 \ell(\theta)}{\partial \beta \partial \beta'} & \frac{\partial^2 \ell(\theta)}{\partial \beta \partial \sigma^2} \\ \frac{\partial^2 \ell(\theta)}{\partial \sigma^2 \partial \beta'} & \frac{\partial^2 \ell(\theta)}{\partial \sigma^2 \partial \sigma^2} \end{pmatrix} = \begin{pmatrix} -\frac{1}{\sigma^2} X'X & -\frac{1}{\sigma^4} X'u \\ -\frac{1}{\sigma^4} u'X & \frac{N}{2\sigma^4} - \frac{u'u}{\sigma^6} \end{pmatrix}$$

The information matrix $\Psi = (\Psi_{ij}) = E(-\mathcal{H}/N)$ is then derived from the Hessian matrix

$$\begin{aligned}\Psi_{11} &= E(-\mathcal{H}_{11}/N) = \frac{1}{\sigma^2} E\left(\frac{1}{N} X' X\right) \equiv \frac{1}{\sigma^2} M_{xx}, \quad \text{say} \\ \Psi_{12} &= \Psi_{21}' = E(-\mathcal{H}_{12}/N) = \frac{1}{\sigma^2} E\left(\frac{1}{N} X' u\right) = \frac{1}{N\sigma^2} \sum_{t=1}^N E(X_t' u_t) = 0 \\ \Psi_{22} &= E(-\mathcal{H}_{22}/N) = -\frac{1}{2\sigma^4} + \frac{1}{\sigma^6} \frac{Eu'u}{N} = -\frac{1}{2\sigma^4} + \frac{1}{\sigma^6} \frac{N\sigma^2}{N} = \frac{1}{2\sigma^4}\end{aligned}$$

Therefore, the asymptotic covariance matrix $V = (N\Psi)^{-1}$ of the MLE is

$$\text{cov}\begin{pmatrix} \hat{\beta} \\ \hat{\sigma}^2 \end{pmatrix} = \begin{pmatrix} \sigma^2 (X'X)^{-1} & 0 \\ 0 & \frac{2\sigma^4}{N} \end{pmatrix}$$

and its estimate is obtained by substituting the MLE of σ^2 .

We noted earlier that the inverse of the negative Hessian matrix, evaluated at the MLE, is often used as the estimate of the covariance matrix. It is easy to see that this gives exactly the same covariance matrix as above, by noting that $X'\hat{u} = 0$ and $\hat{u}'\hat{u} = N\hat{\sigma}^2$.

We showed that the information matrix Ψ can also be computed from

$$\Psi = E \frac{1}{N} \sum_{t=1}^N [g_t(\theta) g_t(\theta)'] = \text{plim} \frac{1}{N} \sum_{t=1}^N [g_t(\theta) g_t(\theta)']$$

where $g_t(\theta) = \partial \ell_t(\theta) / \partial \theta$, the score of a single observation log-likelihood function. Derivatives of ℓ_t with respect to β and σ^2 are

$$\begin{aligned}\frac{\partial \ell_t}{\partial \beta} &= \frac{1}{\sigma^2} (X_t' (y_t - X_t \beta)) = \frac{1}{N} X_t' u_t \\ \frac{\partial \ell_t}{\partial \sigma^2} &= -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4} (y_t - X_t \beta)^2 = -\frac{1}{2\sigma^2} + \frac{u_t^2}{2\sigma^4}\end{aligned}$$

Therefore,

$$\begin{aligned}\Psi_{11} &= \text{plim} \frac{1}{N} \sum \frac{1}{\sigma^4} X_t' u_t u_t' X_t = \frac{1}{\sigma^2} \text{plim} \frac{1}{N} \sum X_t' X_t = \frac{1}{\sigma^2} \text{plim} \frac{X'X}{N} \\ \Psi_{12} &= \text{plim} \frac{1}{N} \sum \frac{1}{\sigma^2} X_t' u_t \left(-\frac{1}{2\sigma^2} + \frac{u_t^2}{2\sigma^4} \right) = -\frac{1}{2\sigma^4} \text{plim} \frac{1}{N} \sum X_t' u_t + \frac{1}{2\sigma^6} \text{plim} \frac{1}{N} \sum X_t' u_t^3 = 0\end{aligned}$$

$$\begin{aligned}
\Psi_{22} &= \text{plim} \frac{1}{N} \sum \left(-\frac{1}{2\sigma^2} + \frac{u_t^2}{2\sigma^4} \right)^2 \\
&= \text{plim} \frac{1}{N} \sum \frac{1}{4\sigma^4} + \frac{1}{4\sigma^8} \left(\text{plim} \frac{1}{N} \sum u_t^4 \right) - \frac{1}{2\sigma^6} \left(\text{plim} \frac{1}{N} \sum u_t^2 \right) \\
&= \frac{1}{4\sigma^4} + \frac{3\sigma^4}{4\sigma^8} - \frac{\sigma^2}{2\sigma^6} = \frac{1}{2\sigma^4}
\end{aligned}$$

where we used the fact that $E(u_t^3)=0$ for a normal u_t with $E(u_t)=0$, and the following relationships

$$\begin{aligned}
\text{plim} \frac{1}{N} \sum X_t' u_t &= E(X_t' u_t) = 0 \\
\text{plim} \frac{1}{N} \sum X_t' u_t^3 &= E(X_t' u_t^3) = 0 \\
\text{plim} \frac{1}{N} \sum u_t^4 &= E(u_t^4) = 3\sigma^4 \\
\text{plim} \frac{1}{N} \sum u_t^2 &= E(u_t^2) = \sigma^2
\end{aligned}$$

Restricted MLE of a Standard Linear Regression Model

Consider a standard linear regression model

$$y_t = X_t \beta + u_t \quad t=1,2,\dots,N, \quad u_t \sim NID(0, \sigma^2), \quad E(u_t | X_t) = 0$$

We wish to estimate the parameters subject a set of J linearly independent linear restrictions

$$h(\theta) = R\beta - q = 0$$

Taking the derivatives of the Lagrange equation

$$\mathcal{L} = \ell(y; X, \theta) - \lambda' (R\beta - q)$$

we have the first order conditions

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \beta} &= \frac{1}{\sigma^2} (X'y - X'X\tilde{\beta}) - R'\tilde{\lambda} = 0 \\ \frac{\partial \mathcal{L}}{\partial \sigma^2} &= -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4} (y - X\tilde{\beta})'(y - X\tilde{\beta}) = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= - (R\tilde{\beta} - q) = 0 \end{aligned}$$

The second equation gives the estimator of σ^2

$$\tilde{\sigma}^2 = (y - X\tilde{\beta})'(y - X\tilde{\beta})/N = \tilde{u}'\tilde{u}/N$$

and the restricted MLE of β and λ are derived from

$$\begin{pmatrix} \tilde{\beta} \\ \tilde{\lambda} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sigma^2} X'X & R' \\ R & 0 \end{pmatrix}^{-1} \begin{pmatrix} \frac{1}{\sigma^2} X'y \\ q \end{pmatrix} = \begin{pmatrix} \hat{\beta} - VR'S^{-1}(R\hat{\beta} - q) \\ S^{-1}(R\hat{\beta} - q) \end{pmatrix}$$

where $\hat{\beta} = (X'X)^{-1}X'y$ is the unrestricted estimator of β , and $V = \sigma^2(X'X)^{-1}$ is the covariance matrix of the unrestricted OLSE, and $S = RVR'$. These results indicate that

- (i) the restricted MLE $\tilde{\beta}$ is the same as the restricted OLSE of β ,
- (ii) the MLE of the Lagrange multiplier is different from its least squares estimator, which is

$$\tilde{\lambda}_{OLS} = \sigma^2 S^{-1}(R\hat{\beta} - q),$$

- (iii) the restricted MLE of σ^2 is different from its restricted least squares estimator $\tilde{u}'\tilde{u}/(N-K+J)$.

The asymptotic covariance matrix of the estimators are easily derived by noting that

$$\frac{\partial h(\theta)}{\partial \theta} = \begin{pmatrix} R' \\ 0 \end{pmatrix} \quad \Psi^{-1} = \begin{pmatrix} \sigma^2 M_{xx}^{-1} & 0 \\ 0 & 2\sigma^4 \end{pmatrix}$$

where $M_{xx} = E(X'X/N)$. This gives

$$\text{cov} \begin{pmatrix} \sqrt{N}(\tilde{\theta} - \theta_0) \\ \frac{1}{\sqrt{N}} \tilde{\lambda} \end{pmatrix} = \begin{pmatrix} \sigma^2 M_{xx}^{-1} - \sigma^2 M_{xx}^{-1} R' (\sigma^2 R M_{xx}^{-1} R')^{-1} \sigma^2 R M_{xx}^{-1} & 0 & 0 \\ 0 & 2\sigma^4 & 0 \\ 0 & 0 & (\sigma^2 R M_{xx}^{-1} R')^{-1} \end{pmatrix}$$

so that

$$\text{cov} \begin{pmatrix} \tilde{\beta} \\ \tilde{\sigma}^2 \\ \tilde{\lambda} \end{pmatrix} = \begin{pmatrix} V - VR'S^{-1}RV & 0 & 0 \\ 0 & 2\sigma^4/N & 0 \\ 0 & 0 & S^{-1} \end{pmatrix}$$

Estimators of β , σ^2 and λ are asymptotically independent.

Substituting these estimators into the log likelihood function, it is straightforward to verify that the maximum value of the log likelihood is

$$\ell(\tilde{\theta}) = -\frac{N}{2}\ln(2\pi) - \frac{N}{2}\ln(\tilde{\sigma}^2) - \frac{N}{2}$$

 Homework #3

(1) A continuous double exponential distribution with a location parameter μ and the scale parameter β has a pdf

$$f(x; \mu, \beta) = \frac{1}{2\beta} \exp\left(-\frac{|x - \mu|}{\beta}\right), \quad -\infty < \mu < \infty, \quad \beta > 0$$

where μ is the mean and β is the scale parameter. The variance of this random variable is $2\beta^2$. Find the MLE of μ and β .

(2) A continuous uniform distribution $U(\alpha, \beta)$ has a pdf

$$f(x; \alpha, \beta) = \begin{cases} \frac{1}{(\beta - \alpha)} & \text{if } \alpha \leq x \leq \beta \\ 0 & \text{otherwise} \end{cases}$$

Specify the likelihood function of a random sample $x_1, x_2, \dots, x_n$, and find the MLE of α and β . Are they consistent estimators? Why or why not?

(3) A Box-Cox model is a nonlinear in parameters and it is specified as

$$y_t(\gamma) = X_t \beta + u_t \quad t=1, 2, \dots, N$$

where

$$y_t(\gamma) = \begin{cases} (y_t^\gamma - 1)/\gamma & \text{if } \gamma \neq 0 \\ \ln y_t & \text{if } \gamma = 0 \end{cases}$$

Assume that $u_t \sim NID(0, \sigma^2)$ and $E(u_t X_t) = 0$. y_t must be strictly positive for this transformation to be defined for all γ .

(a) noting that y_t is monotonic transformation of u_t with Jacobian $J_t = \partial u_t / \partial y_t = y_t^{\gamma-1}$, verify the pdf of y_t

$$f(y_t; X_t, \beta, \sigma^2, \gamma) = (2\pi \sigma^2)^{-1/2} y_t^{\gamma-1} \exp\left(-[y_t(\gamma) - X_t \beta]^2 / (2\sigma^2)\right)$$

(b) Specify the log likelihood function and find the expressions for the gradient vector, i.e., derivatives of the log likelihood function with respect to parameters β , σ^2 and γ .

(c) The gradient vectors are nonlinear in parameters. Find the solutions for β and σ^2 for a given γ , and then substitute them into the log likelihood function to derive the **concentrated log-likelihood function**:

$$\ell(y; X, \beta, \sigma^2, \gamma) = -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\hat{\sigma}^2(\gamma)) + (\gamma - 1) \sum_{t=1}^N \ln y_t - \frac{N}{2}$$

where $\hat{\sigma}^2(\gamma)$ is the solution for σ^2 .

(d) Suggest a procedure to find the solutions of the first order conditions.

(e) Derive the Hessian matrix, and then show that the covariance matrix of $\hat{\theta}$ can be computed by

$$V = \text{cov} \begin{pmatrix} \hat{\beta} \\ \hat{\sigma}^2 \\ \hat{\gamma} \end{pmatrix} = \sigma^2 \begin{pmatrix} X'X & 0 & -X'z \\ 0 & \frac{N}{2\sigma^2} & -\frac{u'z}{\sigma^2} \\ -z'X & -\frac{z'u}{\sigma^2} & w'u + z'z \end{pmatrix}^{-1}$$

where

$$z_t = \frac{\partial[(y_t^\gamma - 1)/\gamma]}{\partial \gamma} = \frac{\gamma y_t^\gamma \ln y_t - y_t^\gamma + 1}{\gamma^2}$$

$$w_t = \frac{\partial z_t}{\partial \gamma} = \frac{y_t^\gamma (\ln y_t)^2 - 2z_t}{\gamma}$$

Tests of Linear Restrictions in a Standard Linear Regression Model

Model: $y_t = X_t \beta + u_t$, $t=1,2,\dots,N$ and $u_t \sim NID(0, \sigma^2)$, $E(u_t X_t) = 0$

Null Hypothesis: $h(\theta) = R\beta - q = 0$, where R is a $J \times K$ matrix of rank J .

Unrestricted estimators: $\hat{\beta} = (X'X)^{-1}X'y$, $\hat{u} = y - X\hat{\beta}$, $\hat{\sigma}^2 = \hat{u}'\hat{u}/N$

Restricted estimators: $\tilde{\beta} = \hat{\beta} - (X'X)^{-1}R'S^{-1}(R\hat{\beta} - q)$, $\tilde{u} = y - X\tilde{\beta}$, $\tilde{\sigma}^2 = \tilde{u}'\tilde{u}/N$

[A] Likelihood Ratio Test

We showed the unrestricted and restricted maximum likelihoods are

$$\ell(y; X, \hat{\theta}) = -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\hat{\sigma}^2) - \frac{N}{2}$$

$$\ell(y; X, \tilde{\theta}) = -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\tilde{\sigma}^2) - \frac{N}{2}$$

The LR test statistic is given by

$$LRT = 2[\ell(y; \hat{\theta}) - \ell(y; \tilde{\theta})] = N \ln(\tilde{\sigma}^2 / \hat{\sigma}^2) \xrightarrow{d} \chi^2(J) \text{ under } H_0$$

Remarks: We may linearize the LRT by taking a Taylor expansion of the logarithmic function around 1 (same estimates of variances, i.e., under H_0):

$$\begin{aligned} LRT &= N \ln(\tilde{\sigma}^2 / \hat{\sigma}^2) = N \ln(z) \approx N \ln(1) + N \frac{\partial \ln(1)}{\partial z} (z - 1) \\ &= N \frac{\tilde{\sigma}^2 - \hat{\sigma}^2}{\hat{\sigma}^2} = N \frac{SSR_r - SSR_u}{SSR_u} \end{aligned}$$

This shows that the LRT is equivalent to comparing restricted and unrestricted SSR's. As we will show later, the finite sample F -test takes this form.

[B] Wald Test

$$\begin{aligned} WT &\equiv N[h(\hat{\theta})]' [H' \Psi(\hat{\theta})^{-1} H]^{-1} [h(\hat{\theta})] \\ &= [h(\hat{\theta})]' [H' [N \Psi(\hat{\theta})]^{-1} H]^{-1} [h(\hat{\theta})] \xrightarrow{d} \chi^2(J) \text{ under } H_0 \end{aligned}$$

where

$$h(\hat{\theta}) = R\hat{\beta} - q$$

$$H(\hat{\theta}) = \frac{\partial h(\hat{\theta})}{\partial \theta} = \begin{pmatrix} R' \\ 0 \end{pmatrix}, \quad (N\Psi(\hat{\theta}))^{-1} = \begin{pmatrix} \hat{\sigma}^2(X'X)^{-1} & 0 \\ 0 & \frac{2\hat{\sigma}^4}{N} \end{pmatrix}$$

$$H' (N\Psi(\hat{\theta}))^{-1} H = \hat{\sigma}^2 R(X'X)^{-1} R' \equiv \hat{S}$$

Hence,

$$WT = (R\hat{\beta} - q)' \hat{S}^{-1} (R\hat{\beta} - q) \xrightarrow{d} \chi^2(J) \text{ under } H_0$$

Recall that $S = \sigma^2 R(X'X)^{-1} R'$ is the covariance matrix of $R\beta$ and $\hat{S}$ is its estimator. Hence, WT is just the quadratic form of a normal random vector $R\hat{\beta} - q$.

[C] Lagrange Multiplier Test

(i) Using the estimate of the Lagrange multiplier $\tilde{\lambda} = [\tilde{\sigma}^2 S]^{-1} (R\hat{\beta} - q)$

$$\begin{aligned} LMT &= \frac{1}{N} \tilde{\lambda}' (H' [\Psi(\tilde{\theta})]^{-1} H) \tilde{\lambda} = \tilde{\lambda}' (H' [N\Psi(\tilde{\theta})]^{-1} H) \tilde{\lambda} \\ &= \tilde{\lambda}' (\tilde{\sigma}^2 S) \tilde{\lambda} = (R\hat{\beta} - q)' (\tilde{\sigma}^2 S)^{-1} (R\hat{\beta} - q) \end{aligned}$$

which is the same as the Wald test, except for the estimate of σ^2 .

(ii) If we use the LMT based on the gradient vector

$$LMT = \left(\frac{1}{\sqrt{N}} \frac{\partial \ell(y; \tilde{\theta})}{\partial \theta} \right)' \Psi^{-1} \left(\frac{1}{\sqrt{N}} \frac{\partial \ell(y; \tilde{\theta})}{\partial \theta} \right) = g(\tilde{\theta})' (N\Psi(\tilde{\theta}))^{-1} g(\tilde{\theta})$$

where $g(\tilde{\theta}) \equiv \partial \ell(y; \tilde{\theta}) / \partial \theta$. From the previous results, we have

$$\begin{aligned} g(\tilde{\beta}) &= \frac{\partial \ell(y; X, \tilde{\theta})}{\partial \beta} = \frac{X'y - X'X\tilde{\beta}}{\tilde{\sigma}^2} = \frac{X'\tilde{u}}{\tilde{\sigma}^2} \\ g(\tilde{\sigma}^2) &= \frac{\partial \ell(y; X, \tilde{\theta})}{\partial \sigma^2} = -\frac{N}{2\tilde{\sigma}^2} + \frac{\tilde{u}'\tilde{u}}{2\tilde{\sigma}^4} = 0 \quad \because \tilde{\sigma}^2 = \frac{\tilde{u}'\tilde{u}}{N} \end{aligned}$$

This gives

$$g(\tilde{\theta}) = \begin{pmatrix} g(\tilde{\beta}) \\ g(\tilde{\sigma}^2) \end{pmatrix} = \begin{pmatrix} \frac{X'\tilde{u}}{\tilde{\sigma}^2} \\ 0 \end{pmatrix}$$

$$\begin{aligned}
LMT &= g(\tilde{\theta})' [N\Psi(\tilde{\theta})]^{-1} g(\tilde{\theta}) = \begin{pmatrix} \frac{\tilde{u}'X}{\tilde{\sigma}^2} & 0 \end{pmatrix} \begin{pmatrix} \tilde{\sigma}^2(X'X)^{-1} & 0 \\ 0 & \frac{2\tilde{\sigma}^2}{N} \end{pmatrix} \begin{pmatrix} \frac{X'\tilde{u}}{\tilde{\sigma}^2} \\ 0 \end{pmatrix} \\
&= \frac{1}{\tilde{\sigma}^2} \tilde{u}'X(X'X)^{-1}X'\tilde{u} \xrightarrow{d} \chi^2(J) \text{ under } H_0
\end{aligned}$$

Remark: Though this expression appears to be different from the expression of the test statistic in (i), they are identical. To verify this, recall

$$\text{Unrestricted estimators: } \hat{\beta} = (X'X)^{-1}X'y, \quad \hat{u} = y - X\hat{\beta}, \quad \hat{\sigma}^2 = \hat{u}'\hat{u}/N$$

$$\text{Restricted estimators: } \tilde{\beta} = \hat{\beta} - (X'X)^{-1}R'S^{-1}(R\hat{\beta} - q), \quad \tilde{u} = y - X\tilde{\beta}, \quad \tilde{\sigma}^2 = \tilde{u}'\tilde{u}/N$$

From these, we have

$$\tilde{u} = y - X\tilde{\beta} = \hat{u} + X(X'X)^{-1}R'S^{-1}(R\hat{\beta} - q)$$

which gives

$$X'\tilde{u} = X'\hat{u} + X'X(X'X)^{-1}R'S^{-1}(R\hat{\beta} - q) = R'S^{-1}(R\hat{\beta} - q)$$

where we used the fact $X'\hat{u} = 0$. Therefore, the LMT based on the gradient vector becomes

$$\begin{aligned}
LMT &= \frac{1}{\tilde{\sigma}^2} \tilde{u}'X(X'X)^{-1}X'\tilde{u} = \frac{1}{\tilde{\sigma}^2} (R\hat{\beta} - q)'S^{-1}R(X'X)^{-1}R'S^{-1}(R\hat{\beta} - q) \\
&= (R\hat{\beta} - q)'[\tilde{\sigma}^2 S]^{-1}(R\hat{\beta} - q)
\end{aligned}$$

where the last equality is due to $S = R(X'X)^{-1}R'$. This expression is the same as the LMT derived from the estimator of the Lagrange multiplier $\tilde{\lambda}$.

Remark: Substituting $\tilde{\sigma}^2$ with $\tilde{\sigma}^2 = \tilde{u}'\tilde{u}/N$, the LMT in (ii) can be written as,

$$LMT = N \frac{\tilde{u}'X(X'X)^{-1}X'\tilde{u}}{\tilde{u}'\tilde{u}} = N \frac{[\tilde{u}'X(X'X)^{-1}X'] [X(X'X)^{-1}X'\tilde{u}]}{\tilde{u}'\tilde{u}} = N \frac{\hat{\tilde{u}}'\hat{\tilde{u}}}{\tilde{u}'\tilde{u}} = NR_0^2$$

where $\hat{\tilde{u}} = X(X'X)^{-1}X'\tilde{u}$ is the predicted value of a linear regression of $\tilde{u}$ on X : $\tilde{u} = X\delta + \epsilon$. R_0^2 is the measure of goodness-of-fit with a modification that the denominator is computed without adjusting the mean of $\tilde{u}$. Recall that the usual definition of R^2 uses $(\tilde{u} - \bar{\tilde{u}})'(\tilde{u} - \bar{\tilde{u}})$ as the denominator.

Tests in the Box-Cox Model

Model:

$$y_t(\gamma) \equiv \frac{y_t^\gamma - 1}{\gamma} = X_t' \beta + u_t, \quad t=1,2,\dots,N, \quad u_t \sim NID(0, \sigma^2), \quad E(X_t' u_t) = 0$$

$$\theta = \begin{pmatrix} \beta \\ \sigma^2 \\ \gamma \end{pmatrix}$$

Hypothesis: $H_0: \gamma = c$ vs $H_1: \gamma \neq c$

Restriction: $h(\theta) = \gamma - c = R\theta - c = 0$, $R = (0, 0, 1) \Rightarrow H = \partial h(\theta) / \partial \theta = R'$

[A] Likelihood Ratio Test

$$LRT = 2(\ell(y; \hat{\theta}) - \ell(y; \tilde{\theta})) = N \ln(\tilde{\sigma}^2 / \hat{\sigma}^2) \sim \chi^2(J) \text{ under } H_0$$

$$\approx N \frac{\tilde{\sigma}^2 - \hat{\sigma}^2}{\hat{\sigma}^2} = N \frac{SSR_r - SSR_u}{SSR_u}$$

[B] Wald Test

$$WT \equiv N[h(\hat{\theta})]' [H' \Psi(\hat{\theta})^{-1} H]^{-1} [h(\hat{\theta})]$$

$$= [h(\hat{\theta})]' [H' [N\Psi(\hat{\theta})]^{-1} H]^{-1} [h(\hat{\theta})] \xrightarrow{d} \chi^2(J) \text{ under } H_0$$

where $h(\hat{\theta}) = \hat{\gamma} - c$ and

$$H(\hat{\theta}) = \frac{\partial h(\hat{\theta})}{\partial \theta} = R' = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad [N\Psi(\hat{\theta})]^{-1} = \hat{\sigma}^2 \begin{pmatrix} X'X & 0 & -X'\hat{z} \\ 0 & \frac{N}{2\hat{\sigma}^2} & -\frac{\hat{u}'\hat{z}}{\hat{\sigma}^2} \\ -\hat{z}'X & -\frac{\hat{z}'\hat{u}}{\hat{\sigma}^2} & \hat{w}'\hat{u} + \hat{z}'\hat{z} \end{pmatrix}^{-1}$$

Further simplification is not practical.

[C] Lagrange Multiplier Test

(i) Using the estimate of the Lagrange multiplier

$$LMT = \frac{1}{N} \tilde{\lambda}' (H' [\Psi(\tilde{\theta})]^{-1} H) \tilde{\lambda} = \tilde{\lambda}' (H' [N\Psi(\tilde{\theta})]^{-1} H) \tilde{\lambda} = (H\tilde{\lambda})' [N\Psi(\tilde{\theta})]^{-1} (H\tilde{\lambda})$$

where $H'=(0,0,1)$, and from the restricted MLE of the Box-Cox model we have

$$\tilde{\lambda} = \sum_{t=1}^N \ln y_t - \frac{\tilde{z}'\tilde{u}}{\tilde{\sigma}^2}, \quad \text{where} \quad \tilde{u} = y(c) - X\tilde{\beta}$$

and $N\Psi(\tilde{\theta})$ is same as in the Wald test, except that we evaluate the matrix by the restricted estimate.

(ii) If we use the LMT based on the gradient vector

$$LMT = \left(\frac{1}{\sqrt{N}} \frac{\partial \ell(y; \tilde{\theta})}{\partial \theta} \right)' \Psi^{-1} \left(\frac{1}{\sqrt{N}} \frac{\partial \ell(y; \tilde{\theta})}{\partial \theta} \right) = g(\tilde{\theta})' [N\Psi(\tilde{\theta})]^{-1} g(\tilde{\theta})$$

From the unrestricted ML estimation of the Box-Cox model, we have

$$g(\tilde{\beta}) = \frac{\partial \ell(y; X, \tilde{\theta})}{\partial \beta} = \frac{X'[y(c) - X\tilde{\beta}]}{\tilde{\sigma}^2} = \frac{X'\tilde{u}}{\tilde{\sigma}^2} = 0 \quad \because \quad \tilde{\beta} = (X'X)^{-1}X'y(c)$$

$$g(\tilde{\sigma}^2) = \frac{\partial \ell(y; X, \tilde{\theta})}{\partial \sigma^2} = -\frac{N}{2\tilde{\sigma}^2} + \frac{\tilde{u}'\tilde{u}}{2\tilde{\sigma}^4} = 0 \quad \because \quad \tilde{\sigma}^2 = \frac{\tilde{u}'\tilde{u}}{N}$$

$$g(\tilde{\gamma}) = \frac{\partial \ell(y; X, \tilde{\theta})}{\partial \gamma} = \sum_{t=1}^N \ln y_t - \frac{\tilde{z}'\tilde{u}}{\tilde{\sigma}^2} = \tilde{\lambda}$$

Therefore,

$$g(\tilde{\theta}) = \begin{pmatrix} g(\tilde{\beta}) \\ g(\tilde{\sigma}^2) \\ g(\tilde{\gamma}) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \tilde{\lambda} \end{pmatrix} = H\tilde{\lambda}$$

and the LMT test statistic is same as the one derived from the distribution of $\tilde{\lambda}$.

Appendix

Proof of asymptotic properties of MLE

Notations: For notational simplicity we use the following notations.

Log likelihood function: $\ell(\boldsymbol{\theta}) = \ell(\mathbf{y}; \boldsymbol{\theta}) = \sum_{t=1}^N \ln f(\mathbf{y}_t; \boldsymbol{\theta})$

Gradient vector: $\mathbf{g}(\boldsymbol{\theta}) = \partial \ell(\boldsymbol{\theta}) / \partial \boldsymbol{\theta}$

Hessian matrix: $\mathcal{H}(\boldsymbol{\theta}) = \frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} = \frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'}$

Information matrix: $\boldsymbol{\Psi} = E \left(-\frac{1}{N} \mathcal{H}(\boldsymbol{\theta}) \right)$

Single observation log likelihood: $\ell_t(\boldsymbol{\theta}) = \ell(\mathbf{y}_t; \boldsymbol{\theta}) = \ln f(\mathbf{y}_t; \boldsymbol{\theta})$

Single observation gradient vector: $\mathbf{g}_t(\boldsymbol{\theta}) = \partial \ell_t(\boldsymbol{\theta}) / \partial \boldsymbol{\theta}$

Single observation Hessian matrix: $\mathcal{H}_t(\boldsymbol{\theta}) = \frac{\partial^2 \ell_t(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} = \frac{\partial \mathbf{g}_t(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'}$

Single observation Information matrix: $\boldsymbol{\Psi} = E[-\mathcal{H}_t(\boldsymbol{\theta})]$

Note that $\ell(\boldsymbol{\theta}) = \sum \ell_t(\boldsymbol{\theta})$, $\mathbf{g}(\boldsymbol{\theta}) = \sum \mathbf{g}_t(\boldsymbol{\theta})$ and $\mathcal{H}(\boldsymbol{\theta}) = \sum \mathcal{H}_t(\boldsymbol{\theta})$. When we evaluate these functions at a specific value of $\boldsymbol{\theta}$, it will be indicated by, for example, $\mathbf{g}(\hat{\boldsymbol{\theta}})$ for $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}$.

The proof of this theorem relies on the following results.

Propositions

Proposition 1: $E[\ell(\boldsymbol{\theta}_0)/N] \geq E[\ell(\boldsymbol{\theta})/N]$ for all $\boldsymbol{\theta} \in \Theta$.

Proposition 2: $\text{plim}_{N \rightarrow \infty} \ell(\boldsymbol{\theta})/N = E[\ell(\boldsymbol{\theta})/N]$ for all $\boldsymbol{\theta} \in \Theta$.

Proposition 3: $E[\mathbf{g}(\boldsymbol{\theta})] = E[\mathbf{g}_t(\boldsymbol{\theta})] = \mathbf{0}$, $\text{cov}[\mathbf{g}(\boldsymbol{\theta})] = N\boldsymbol{\Psi}$, $\text{cov}[\mathbf{g}_t(\boldsymbol{\theta})] = \boldsymbol{\Psi}$

Proposition 4: $\frac{1}{\sqrt{N}} \mathbf{g}(\boldsymbol{\theta}) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{\Psi})$

Proposition 5: $\text{plim}_{N \rightarrow \infty} \left(-\frac{1}{N} \mathcal{H}(\boldsymbol{\theta}) \right) = \boldsymbol{\Psi}$

Proof of Theorem and Propositions

Proof of Consistency:

From Propositions 1 and 2 we have

$$plim[\ell(\theta_0)/N] = E[\ell(\theta_0)/N] \geq E[\ell(\theta)/N] = plim[\ell(\theta)/N]$$

for all θ . In particular, the inequality holds for $\theta = \hat{\theta}$ and hence,

$$plim[\ell(\theta_0)/N] \geq plim[\ell(\hat{\theta})/N]$$

But $\hat{\theta}$ is the MLE for every N , and hence,

$$\ell(\hat{\theta})/N \geq \ell(\theta)/N$$

for all θ (including θ_0) and for all N . Therefore,

$$plim[\ell(\hat{\theta})/N] \geq plim[\ell(\theta_0)/N]$$

These two inequalities imply $plim[\ell(\hat{\theta})/N] = plim[\ell(\theta_0)/N]$. This in turn implies $plim(\hat{\theta}) = \theta_0$ when the model is *asymptotically identified*. The model is asymptotically identified under very general conditions which are satisfied by the regularity conditions (see Rao).

Proof of Asymptotic Normality:

Take the first order Taylor expansion of $g(\hat{\theta})$ around the true value θ_0 :

$$g(\hat{\theta}) = g(\theta_0) + \mathcal{H}(\theta_0)(\hat{\theta} - \theta_0)$$

Since $g(\hat{\theta}) = 0$, dividing the equation with $\sqrt{N}$ and rearranging the terms we can write

$$\sqrt{N}(\hat{\theta} - \theta_0) = \left(-\frac{1}{N} \mathcal{H}(\theta_0) \right)^{-1} \frac{1}{\sqrt{N}} g(\theta_0)$$

The first matrix on the right hand side converges in probability to Ψ by Proposition 5, and the last term converges in distribution to $N(0, \Psi)$ by Proposition 4. Hence, by using the well-known theorem¹, $\sqrt{N}(\hat{\theta} - \theta_0)$ converges in distribution to $\Psi^{-1}Z$, where $Z \sim N(0, \Psi)$. This gives the desired result.

Proof of Proposition 1.

Since $\ln(z)$ is a strictly concave function, we have by the Jensen's inequality $E[\ln(z)] \leq \ln E(z)$ for a random variable z . Take $z = L(y; \theta)/L(y; \theta_0)$, the ratio of the likelihood functions. Then,

$$E[\ell(\theta)/N] - E[\ell(\theta_0)/N] = E[\ln z] \leq \ln E(z) = \ln \int \frac{L(y; \theta)}{L(y; \theta_0)} L(y; \theta_0) dy = \ln(1) = 0$$

¹ Theorem: If $plim(a_n) = c$ for a nonzero constant c and Z_n converges to Z in distribution, then $(a_n Z_n)$ converges to cZ in distribution.

Proof of Proposition 2.

$$\frac{1}{N} \ell(\theta) = \frac{1}{N} \sum_{t=1}^N \ell_t(\theta) \equiv \bar{\ell}(\theta)$$

$\ell_t(\theta) = \ln f(y_t; \theta)$ is an i.i.d. random variable. Hence, by the WLLN, their sample mean $\bar{\ell}(\theta)$ converges to the mean $E \ell_t(\theta) = E \bar{\ell}(\theta)$ in probability.

Proof of Proposition 3. Since $f(y_t; \theta)$ is a density function it must integrate to 1:

$$\int f(y_t; \theta) dy_t = 1$$

Differentiation of both sides of this equation with respect to θ gives²

$$(1) \quad \int g_t(\theta) f(y_t; \theta) dy_t = 0$$

The left hand side is simply the definition of the expected value of $g_t(\theta)$, and hence $E[g_t(\theta)] = 0$.

Differentiating (1) again with respect to θ , we can write

$$\int \mathcal{H}_t(\theta) f(y_t; \theta) dy_t + \int (g_t(\theta) g_t(\theta)') f(y_t; \theta) dy_t = 0$$

The first term is $E[\mathcal{H}_t(\theta)]$, and the second term is $E[g_t(\theta) g_t(\theta)']$, which is the covariance of $g_t(\theta)$ because $E[g_t(\theta)] = 0$. Therefore,

$$\text{cov}(g_t(\theta)) = E[g_t(\theta) g_t(\theta)'] = E[-\mathcal{H}_t(\theta)] = \Psi$$

Since $g_t(\theta)$ is an i.i.d. random variable, we have

$$\begin{aligned} E[g(\theta)] &= \sum_{t=1}^N E[g_t(\theta)] = 0 \\ \text{cov}(g(\theta)) &= \text{cov} \sum_{t=1}^N (g_t(\theta)) = \sum_{t=1}^N \text{cov}(g_t(\theta)) = N\Psi \end{aligned}$$

These results for the entire sample can also be shown by using the same procedure as above. Since likelihood function $L(y; \theta)$ is a joint density function it must integrate to 1:

$$\int L(y; \theta) dy = 1$$

Differentiation of both sides of this equation with respect to θ gives

² We use the relationship $\partial \ln f / \partial \theta = (1/f)(\partial f / \partial \theta)$, which gives $\partial f / \partial \theta = (\partial \ln f / \partial \theta) f$.

$$(2) \quad \int \mathbf{g}(\boldsymbol{\theta}) L(\mathbf{y}; \boldsymbol{\theta}) d\mathbf{y} = \mathbf{0}$$

The left hand side is simply the definition of the expected value of $\mathbf{g}(\boldsymbol{\theta})$ because $L(\mathbf{y}; \boldsymbol{\theta})$ is the joint density of $\mathbf{y}$. Therefore, $E[\mathbf{g}(\boldsymbol{\theta})] = \mathbf{0}$.

Differentiating (2) again with respect to $\boldsymbol{\theta}$, we can write

$$\int \mathcal{H}(\boldsymbol{\theta}) L(\mathbf{y}; \boldsymbol{\theta}) d\mathbf{y} + \int (\mathbf{g}(\boldsymbol{\theta}) \mathbf{g}(\boldsymbol{\theta})') L(\mathbf{y}; \boldsymbol{\theta}) d\mathbf{y} = \mathbf{0}$$

The first term is $E[\mathcal{H}(\boldsymbol{\theta})]$, and the second term is $E[\mathbf{g}(\boldsymbol{\theta}) \mathbf{g}(\boldsymbol{\theta})']$, which is the covariance of $\mathbf{g}(\boldsymbol{\theta})$ because $E[\mathbf{g}(\boldsymbol{\theta})] = \mathbf{0}$. Therefore,

$$\text{cov}(\mathbf{g}(\boldsymbol{\theta})) = E[-\mathcal{H}(\boldsymbol{\theta})] = N\Psi$$

Proof of Proposition 4.

$$\frac{1}{\sqrt{N}} \mathbf{g}(\boldsymbol{\theta}) = \frac{1}{\sqrt{N}} \sum_{t=1}^N \mathbf{g}_t(\boldsymbol{\theta})$$

This converges in distribution to $N(\mathbf{0}, \Psi)$ by the central limit theorem, because $\{\mathbf{g}_t(\boldsymbol{\theta})\}$ is a sequence of i.i.d. random vector with zero mean and covariance matrix Ψ by Proposition 3.

Proof of Proposition 5.

$$\frac{1}{N} \mathcal{H}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{t=1}^N \mathcal{H}_t(\boldsymbol{\theta}) = \bar{\mathcal{H}}_t(\boldsymbol{\theta})$$

Since $\{\mathcal{H}_t(\boldsymbol{\theta})\}$ is a sequence of i.i.d. random matrix, its sample mean $\bar{\mathcal{H}}_t(\boldsymbol{\theta})$ converges in probability to its true mean $E(\mathcal{H}_t) = -\Psi$ by Khinchine's WLLN.

Alternative Covariance Estimators of the MLE

We showed that the MLE has an asymptotic normal distribution

$$\sqrt{N}(\hat{\boldsymbol{\theta}}_N - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \Psi^{-1})$$

We also showed the equivalence of

$$\begin{aligned} \Psi &= E\left(-\frac{1}{N} \mathcal{H}(\boldsymbol{\theta})\right) = E[-\mathcal{H}_t(\boldsymbol{\theta})] = \text{plim}\left(-\frac{1}{N} \mathcal{H}(\boldsymbol{\theta})\right) \\ &= E\left(\frac{1}{N} \mathbf{g}(\boldsymbol{\theta}) \mathbf{g}(\boldsymbol{\theta})'\right) = E[\mathbf{g}_t(\boldsymbol{\theta}) \mathbf{g}_t(\boldsymbol{\theta})'] = \text{plim}\left(\frac{1}{N} \sum_{t=1}^N \mathbf{g}_t(\boldsymbol{\theta}) \mathbf{g}_t(\boldsymbol{\theta})'\right) \end{aligned}$$

Note that the first three expressions require computation of the Hessian matrix, while the last three expressions require only the gradient vector. If the expectations of these forms are functions of the sample size N , then their limit as $N \rightarrow \infty$ is sometimes taken for Ψ .

In practice, we are interested in the MLE $\hat{\theta}$ and its asymptotic distribution. $\hat{\theta}$ is asymptotically distributed as normal with mean θ_0 (true value of θ) and covariance matrix $V \equiv (N\Psi)^{-1}$, where Ψ is evaluated at the MLE $\hat{\theta}$. The choice of Ψ to be used in V depends primarily on the degree of difficulty in computing the expectations or *plim*.

When it is difficult to compute them, their empirical values (matrices without taking expectations or *plim*) are often used for the estimate of V :

$$V = \text{asympt. cov}(\hat{\theta}) = \left(-\mathcal{H}(\hat{\theta})\right)^{-1} \quad \text{or} \quad \left(g(\hat{\theta})g(\hat{\theta})'\right)^{-1} \quad \text{or} \quad \left(\sum_{t=1}^N g_t(\hat{\theta})g_t(\hat{\theta})'\right)^{-1}$$

In particular, the inverse of the negative Hessian matrix is used as the estimate of V in many software.

In some cases, it is easy to compute the expectation or *plim* for some elements in the matrix, and not easy for other elements (see the example of the Box-Cox model below). In such cases, it is generally suggested to use the expected value or *plim* for those elements that can be easily computed, and use the empirical values for other elements.

Appendix

Proof of Asymptotic Distribution of Likelihood Ratio Test Statistic

Theorem. Let $\hat{\theta}$ and $\tilde{\theta}$ be the unrestricted MLE and restricted MLE of θ subject to $h(\theta)=0$, respectively. Let $\ell(y; \hat{\theta})=\ln L(y; \hat{\theta})$ and $\ell(y; \tilde{\theta})=\ln L(y; \tilde{\theta})$ be their log-likelihoods. Under the regularity conditions, the likelihood ratio test statistic

$$LRT \equiv 2 \ln \Lambda = 2[\ell(y; \hat{\theta}) - \ell(y; \tilde{\theta})]$$

has an asymptotic null distribution of a central $\chi^2(J)$, where J is the number of the restrictions, and it has a noncentral χ^2 distribution under the alternative hypothesis.

Proof. Take a second order Taylor's expansion of the restricted likelihood $\ell(y; \tilde{\theta})$ around $\hat{\theta}$:

$$\ell(y; \tilde{\theta}) \approx \ell(y; \hat{\theta}) + \frac{\partial \ell(y; \hat{\theta})}{\partial \theta} (\tilde{\theta} - \hat{\theta}) + \frac{1}{2} (\tilde{\theta} - \hat{\theta})' \frac{\partial^2 \ell(y; \hat{\theta})}{\partial \theta \partial \theta'} (\tilde{\theta} - \hat{\theta})$$

Since $\hat{\theta}$ is the unrestricted MLE, $\partial \ell(y; \hat{\theta}) / \partial \theta = 0$. Hence,

$$\begin{aligned} 2 \ln \Lambda &= 2[\ell(y; \hat{\theta}) - \ell(y; \tilde{\theta})] = -(\tilde{\theta} - \hat{\theta})' \frac{\partial^2 \ell(y; \hat{\theta})}{\partial \theta \partial \theta'} (\tilde{\theta} - \hat{\theta}) \\ &= [\sqrt{N}(\hat{\theta} - \theta_0) - \sqrt{N}(\tilde{\theta} - \theta_0)]' \left(-\frac{1}{N} \frac{\partial^2 \ell(y; \hat{\theta})}{\partial \theta \partial \theta'} \right) [\sqrt{N}(\hat{\theta} - \theta_0) - \sqrt{N}(\tilde{\theta} - \theta_0)] \end{aligned}$$

We have shown before the following results:

$$-\frac{1}{N} \frac{\partial^2 \ell(y; \hat{\theta})}{\partial \theta \partial \theta'} \xrightarrow{P} \Psi$$

$$\sqrt{N}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \Psi^{-1})$$

$$\sqrt{N}(\tilde{\theta} - \theta_0) \xrightarrow{d} N(\tilde{\mu}, P)$$

where

$$\tilde{\mu} \equiv \Psi^{-1} H (H' \Psi^{-1} H)^{-1} h(\theta_0)$$

$$P \equiv \Psi^{-1} - \Psi^{-1} H (H' \Psi^{-1} H)^{-1} H' \Psi^{-1} = P \Psi P$$

Let Z be a standard normal random vector: $Z \sim (0, I)$, and let A be the nonsingular square root matrix of Ψ , i.e., $\Psi = A' A$ and $\Psi^{-1} = A^{-1} A'^{-1}$. Then, $A^{-1} Z \sim N(0, \Psi^{-1})$, and $PA' Z \sim N(0, P)$ because $PA' A P = P \Psi P = P$. Therefore,

$$\sqrt{N}(\hat{\theta} - \theta_0) \stackrel{ad}{\approx} A^{-1} Z$$

$$\sqrt{N}(\tilde{\theta} - \theta_0) \stackrel{ad}{\approx} PA' Z + \tilde{\mu}$$

where $\stackrel{ad}{\sim}$ stands for "asymptotically same distribution". We can now write

$$\begin{aligned}
 2\ln\Lambda &\stackrel{ad}{\sim} [\sqrt{N}(\hat{\theta} - \theta_0) - \sqrt{N}(\tilde{\theta} - \theta_0)]' \Psi [\sqrt{N}(\hat{\theta} - \theta_0) - \sqrt{N}(\tilde{\theta} - \theta_0)] \\
 &\stackrel{ad}{\sim} (A^{-1}Z - PA'Z - \tilde{\mu})' \Psi (A^{-1}Z - PA'Z - \tilde{\mu}) \\
 &= (Z - \tilde{\mu}^*)' (A^{-1} - PA')' \Psi (A^{-1} - PA')(Z - \tilde{\mu}^*) = (Z - \tilde{\mu}^*)' (I - APA')(Z - \tilde{\mu}^*)
 \end{aligned}$$

where $\tilde{\mu}^* = (A^{-1} - PA')^{-1}\tilde{\mu}$.

It is easy to verify that $(I - APA')$ is a symmetric idempotent matrix of rank J , the number of restrictions. Hence, by the well known theorem, $2\ln\Lambda$ has an asymptotic distribution $\chi^2(J, \delta)$, where the noncentrality parameter is given by

$$\delta^2 = (\tilde{\mu}^*)' (I - APA') (\tilde{\mu}^*) = \tilde{\mu}' \Psi \tilde{\mu} = h(\theta_0)' (H' \Psi^{-1} H)^{-1} h(\theta_0)$$

Under the null hypothesis, $\tilde{\mu} = 0$, and $2\ln\Lambda$ has a central $\chi^2(J)$.

Q.E.D.